

Real-time Public Transport Tracking System for Small Cities

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Abstract—Modern urban mobility systems need continuous tracking of transportation services because it has become an essential feature. The arrival time and location of buses remain unknown to many passengers, which creates problems and leads to inefficient travel methods. The paper introduces a complete bus tracking system, which includes real-time tracking capabilities and fleet management functions to enhance public transportation operations. The proposed system uses Node.js and Express for backend development, Socket.IO for real-time communication, and Leaflet maps for location visualization. The system provides three main interfaces: a passenger dashboard for live bus tracking, a driver dashboard for trip status control, and an admin panel for fleet management. The system uses WebSocket technology to transmit real-time bus location updates, which allows users to watch bus movements without needing to refresh their screens. The system contains additional features that include estimated time of arrival (ETA) and passenger capacity indicators, along with route visualization. The implementation demonstrates how real-time communication technologies can enhance the efficiency and usability of public transportation systems.

Index Terms—Real-Time Tracking, Public Transportation, WebSocket Communication, Node.js, Fleet Management, Leaflet Map

I. INTRODUCTION

Cities depend on their public transportation systems to serve the daily commuting needs of their residents. Buses represent one of the most popular and economical options for people who need to travel. Passengers experience difficulties because they cannot determine when buses will arrive at bus stops and where buses currently are. Bus schedules show estimated times but they do not show real-time traffic conditions or any delays that may occur.

The development of internet technologies together with real-time communication systems has enabled the implementation of live vehicle tracking systems which deliver instant location information. The real-time bus tracking system enables passengers who need to travel to check the current position of buses and their expected time of arrival at bus stops.

The project creates a real-time bus tracking system which uses contemporary web development tools to build its fleet management system. The system enables passengers to track buses through a map interface while drivers can update bus operational status and administrators control bus fleet operations. The system achieves real-time updates through WebSocket communication which uses Socket.IO to let the server send data to all connected clients without delay. The project aims to create a basic operational system which shows how real-time communication technologies enhance public transportation system management and monitoring capabilities.

The proposed system shows its essential value through its capacity to simulate multiple buses which operate in a transportation network. The system includes a bus model which shows its unique identification number and route details and its location coordinates and maximum passenger capacity and speed and current operational state. The server uses its automated processes to create updates which display both movement and operational changes of the system. All users who are currently connected to the system receive updates at the same time. The system establishes a complete monitoring system which enables centralized control to track bus conditions and their entire fleet movements.

The implementation of such a system provides several advantages for both passengers and transportation administrators. Passengers receive precise bus location data which enables them to minimize their waiting period at bus stops. Administrators gain the ability to oversee bus operations while they handle fleet management tasks and emergency situations which include breakdowns or service interruptions. The system currently implements simulated data but its architecture permits seamless integration of actual GPS devices which transportation organizations use for monitoring their bus operations.

II. RELATED WORK

A. Maintaining the Integrity of the Specifications

The concept of real-time public transport tracking has gained significant attention in recent years because urban

populations continue to grow and people need better public transportation systems. Researchers have developed solutions that use GPS technology and mobile communication systems and online platforms to improve public transit systems for better user experience and service reliability.

J Babu et al. [1] developed a GPS-based bus tracking system through their research to allow users to track bus locations in real time. The system used location data to send vehicle information from the bus to a central server which users could access through a web platform. The system operated well but it did not include features that allowed users to interact with the system and administrators to manage the system.

G Gayathri et al. [3] conducted a study which investigated a real-time bus tracking system that uses both GPS and GSM technologies for its operation. Passengers could track bus locations through their mobile devices because of their developed system. The system focused on tracking operations yet it lacked a multi-user system which would enable different users to access distinct dashboards designed for administrators and drivers and passengers.

Mane and Khairnar [4] analyzed the feasibility of implementing bus tracking systems using smartphones equipped with GPS capabilities. The study discovered that mobile-based tracking solutions proved to be more affordable than dedicated tracking hardware devices. The study showed practical results yet it failed to provide the necessary features which enable organizations to control multiple buses and routes in actual operational situations.

The authors of the research study [6] implemented a GPS-based real-time vehicle tracking system which aimed to enhance transportation operations according to their study. The system proved its ability to track vehicles in real time however it lacked essential features which modern intelligent transport systems require to manage vehicle fleets and update information for various parties.

The recent studies conducted by Gogri et al.[10] and Thomas et al. [9] which were published after the initial research, focus on two main topics which are immediate data evaluation and advanced tracking systems. The systems developed new features which enhanced their location precision and achieved faster data transfer through contemporary web design methods. The organization still needed to develop unified management systems which would enable supervisors to control vehicle fleets while they observed operations throughout the entire fleet.

The researchers et al.[8] developed a smart bus tracking system which used location-aware services and QR code technology for enhancing passenger interaction according to their research. The novel method developed by them which offered multiple advantages to passengers, remained limited because it did not include management tools for system operation or capabilities to control drivers.

The studies by Bessa et al.[13] and Nguyen et al. [14] examined how public transport systems use GPS trajectory data to conduct their analysis. The two studies made important contributions which advanced knowledge about route opti-

mization and anomaly detection. The research presented an analytical study which failed to demonstrate a full real-time system functioning.

The research by Ganesh et al. [16] and his colleagues developed a passenger information system which delivers current bus arrival and departure information to users. The system made user experience improvements through its design but it failed to include elements which would allow drivers to provide input and administrators to monitor operations.

Most existing systems which researchers studied in the literature survey concentrate their efforts on either tracking or passenger information services. Only a few systems exist which deliver complete solutions that combine real-time tracking capabilities with dedicated role-based dashboards designed for use by administrators and drivers and passengers. Many existing systems fail to provide scalable solutions which enable multiple users to access real-time data while managing their fleets in a flexible way.

The proposed system creates a new solution which unites bus tracking systems with three separate dashboard interfaces for administrators and drivers and passengers. The admin panel enables fleet management operations such as adding or removing buses, while the driver dashboard allows real-time status updates including trip start, end, and breakdown reporting. Live tracking information from the passenger interface allows users to experience better reliability and user experience. The system builds its interactive platform through component integration which uses contemporary technologies for real-time communication, making it suitable for small and medium-sized cities.

III. REAL-TIME PUBLIC TRANSPORT TRACKING SYSTEM ARCHITECTURE

The system shown in below Figure 1 is a real-time communication framework that consists of multiple interconnected modules for tracking and managing public transport efficiently:

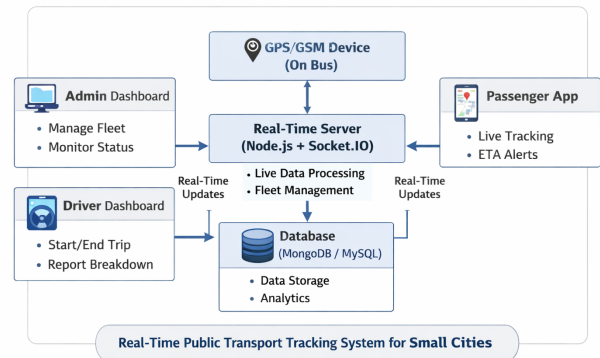


Fig. III.1. Block diagram showing the functional flow of the web application.

A. OVERALL SYSTEM DESIGN

As shown in Fig. III.1, the block diagram represents the overall workflow of the system. The design of the Real-Time Public Transport Tracking System functions as a web-based distributed monitoring system that enables real-time vehicle tracking and fleet management and operational control through its centralized real-time communication feature.

The system architecture unites interactive map visualization and administrative monitoring tools and driver controls and backend event processing into a single transport management system. The architectural design shown in Fig. 1 operates through a client-server communication system which enables all user modules to connect with the central real-time server that was constructed using Node.js and Socket.IO technology. The system is structured into five primary modules:

- o Real-Time Processing Server
- o Passenger Tracking Interface
- o Driver Dashboard
- o Administrative Dashboard

The system modular design provides three main advantages of scaling capabilities and system upkeep requirements and effective data sharing between different transport system interfaces.

B. REAL-TIME PROCESSING SERVER

The real-time processing server functions as the main processing unit for the proposed system because it manages all communication activities between client systems. The system uses Node.js and Socket.IO to create an asynchronous environment which supports event-driven communication needed for actual real-time applications with minimal latency.

The server performs the following functions:

- o The system demonstrates moving vehicles through continuous updates of their transportation locations. The system sends current location information to all clients who are connected. The system handles all fleet management requests which administrators submit. The system processes all operational events which drivers create. The system ensures that all dashboards receive complete data synchronization.

- o The system updates vehicle location data at regular intervals to create an academic simulation which shows actual vehicle movement. The simulated movement enables testing of live transport tracking functionality under conditions which closely match actual operating environments.

- o Socket.IO provides continuous WebSocket connections between the server and client interfaces which results in faster communication and better update performance than traditional request-response systems.

C. PASSENGER TRACKING INTERFACE

The passenger tracking interface provides commuters with real-time visibility into transport movement through an interactive digital map. The system uses Leaflet.js together with OpenStreetMap for displaying real-time bus position and travel path data to users.

The system obtains its location information through simulated data which the backend generates and displays on the map at continually updated intervals.

The passenger interface contains its main features through which users can:

- o Live tracking of moving buses on the map
- o Dynamic marker movement visualization
- o Route path representation using polyline mapping
- o Bus stop indication through static markers
- o Estimated Time of Arrival (ETA) calculation
- o Capacity monitoring through occupancy indicators

The module enables users to watch bus locations together with their arrivals which benefits passengers to a greater extent.

D. DRIVER DASHBOARD

The driver dashboard provides operational controls for transport personnel and allows drivers to communicate trip-related events directly to the centralized server. The dashboard includes control actions which enable users to:

- o Start Trip
- o End Trip
- o Report Vehicle Breakdown

The system sends these actions to the backend server through Socket.IO events which then distribute the changes to all active dashboards. The module enhances communication between field operators and system administrators while providing operational information to all parties involved. The dashboard shows the active trip status of the selected vehicle which enables drivers to check their current operational state.

E. ADMINISTRATIVE PANEL

The administrative dashboard functions as the supervisory interface for transport management authorities. The system enables centralized control over fleet operations while monitoring all active vehicles throughout the system.

The administrator is provided with tools for:

- o Adding new buses to the transport fleet
- o Removing inactive buses
- o Monitoring fleet status in real time
- o Viewing occupancy/capacity statistics
- o Analyzing average transport utilization

The dashboard improves fleet management efficiency by offering a centralized interface for overseeing transport operations.

F. SYSTEM WORKFLOW

The proposed system workflow starts when the backend simulation engine produces regular updates which show the current location of moving transport vehicles. The real-time server processes these updates before sending them to all connected interfaces via WebSocket communication.

- o Passenger interfaces receive these updates and reflect them through map-based movement visualization. At the same time, administrative and driver dashboards receive updates about operational information which includes fleet status and trip

activity. o Administrative and driver-generated commands are sent back to the server and synchronized across all modules to maintain consistency throughout the platform. o The bi-directional communication system allows all stakeholders in the transport management environment to work together without any difficulties.

IV. METHODOLOGY

A. VEHICLE MOVEMENT SIMULATION

The system implements a simulation-based method to show how vehicles move through the transportation system. The project needs to show academic results so it does not use actual tracking devices for its demonstration. The backend system simulates vehicle movement by updating their latitude and longitude coordinates at regular intervals throughout the entire route. The system provides users with ongoing bus trackable movements that show real-time bus routes through the interactive map feature.

B. REAL-TIME SYNCHRONIZATION

The system uses Socket.IO to enable event-based data sharing between its backend system and client user interfaces. The server sends new vehicle location information to all active users immediately after any changes to vehicle tracking data or service operational information take place. The system provides real-time updates to passenger driver and administrative dashboards while users do not need to refresh their displays manually.

C. ETA COMPUTATION

The system calculates bus arrival time by measuring the travel time needed for the bus to reach its destination from its current location based on its speed and distance. The estimated arrival time is computed using:

$ETA = (Distance / speed) * 60$ The system shows estimated time of arrival to passengers through the dashboard which displays arrival times in minute intervals for passenger tracking purpose.

D. OCCUPANCY CLASSIFICATION

The system uses a capacity-based classification model to show how many passengers are currently present. The occupancy percentage of each vehicle is grouped into three categories:

- o Low Crowd: Below 40 Percent
- o Medium Crowd: Between 40 and 80 Percent
- o High Crowd: Above 80 Percent

Users can easily read the different occupancy levels through color-coded visual markers which show different levels of occupancy.

V. . RESULTS AND DISCUSSION

Table I summarizes the implementation parameters and performance characteristics of the proposed system.

The study compares traditional HTTP polling with WebSocket communication through Figure V.1. The WebSocket

TABLE I
SYSTEM PERFORMANCE AND IMPLEMENTATION PARAMETERS

Parameter	Value
Update Interval	3 Seconds
Number of Simulated Buses	3–50
Communication Protocol	WebSocket
Average Response Time	< 200 ms
Map Rendering Tool	Leaflet.js
Backend Framework	Node.js

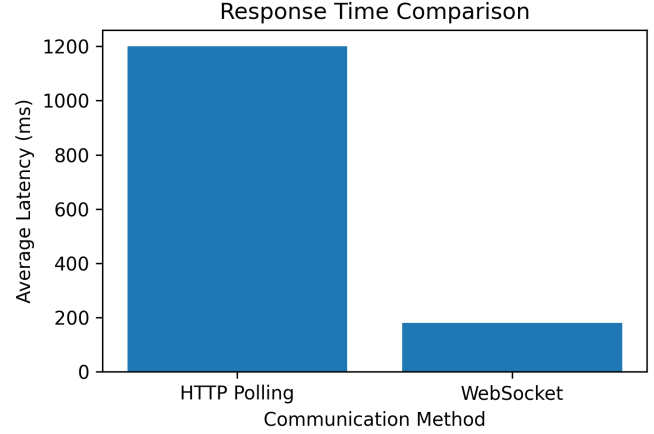


Fig. V.1. Response time comparison between HTTP polling and WebSocket communication.

protocol demonstrates superior performance to other methods because it achieves faster response times which make it ideal for real-time applications that require constant updates.

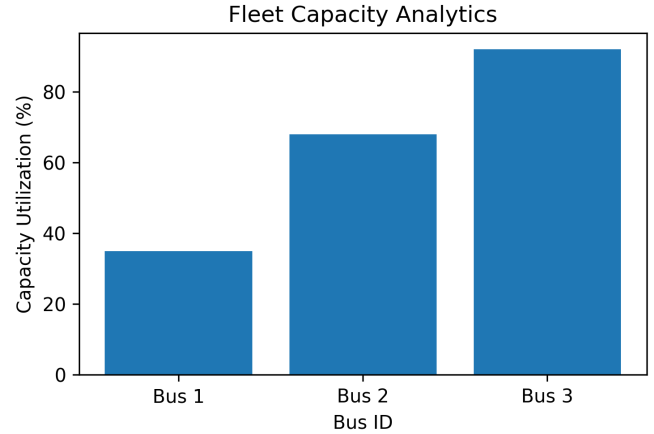


Fig. V.2. Fleet occupancy analysis of simulated buses.

The simulation results in Figure V.2 display how the transport fleet occupancy patterns throughout the simulation. The analysis permits evaluation of passenger crowd density which helps users make informed decisions about boarding buses based on their current capacity.

Figure V.3 shows how the fleet distributes its operational activities across all vehicles. The administrative dashboard con-

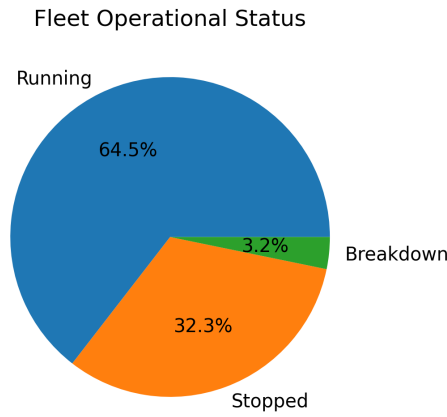


Fig. V.3. Distribution of fleet operational statuses.

tinuously monitors vehicle conditions and categorizes buses into running, stopped, and breakdown states for effective fleet supervision.

A. PRACTICAL IMPLICATIONS

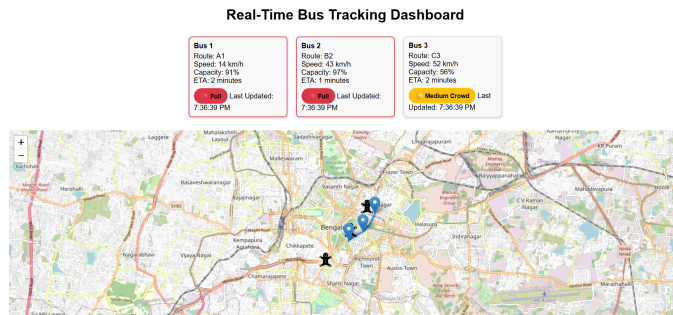


Fig. V.4. Passenger dashboard displaying real-time bus tracking and route information.

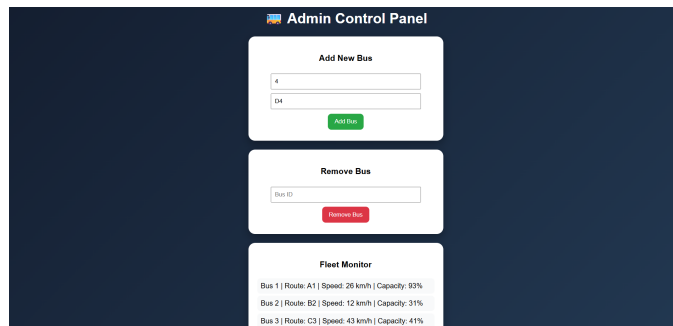


Fig. V.5. Administrative dashboard for centralized fleet monitoring and management.

VI. CONCLUSION

The real-time public transport tracking system which has been proposed provides a scalable and efficient solution that

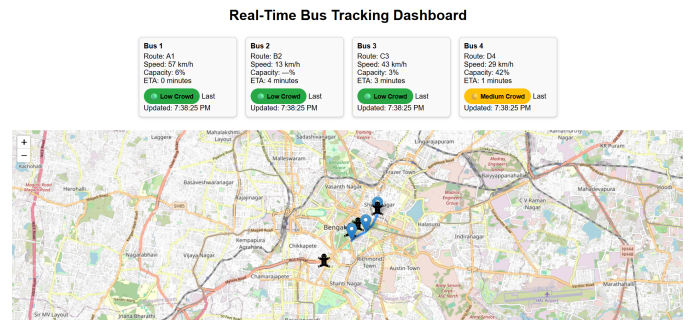


Fig. V.6. Passenger interface showing live bus movement on the interactive route map.

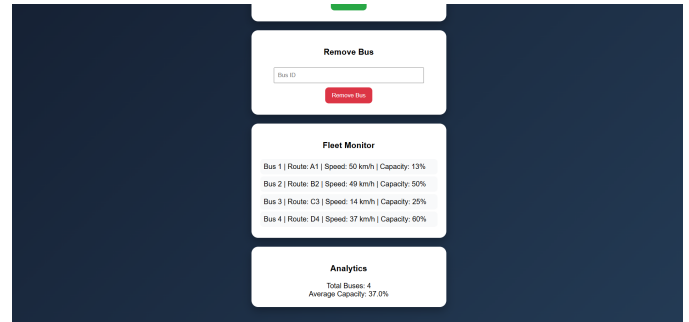


Fig. V.7. Fleet monitoring dashboard displaying active bus operational statistics.

enables better public transportation monitoring and management operations in small growing cities. The system uses Web-Socket technology for real-time communication and Leaflet.js for interactive map visualization which enables passengers drivers and administrators to monitor synchronized transport information through their respective interfaces. The framework which scientists created demonstrates its capability to track vehicles in real-time calculate arrival times and monitor fleet occupancy while providing operational control through separate dashboards for admin and driver users.

The simulation-based implementation demonstrates that the proposed architecture functions effectively in academic settings because it does not need physical tracking equipment yet it preserves realistic simulation of transport movements and operational processes. When users test the system through experimentation they find that WebSocket communication enables faster updates which result in less delay time compared to standard polling methods thus making the system run better.

The proposed platform delivers a digital transport management solution which users find easy to use and it establishes a base for developing future smart transportation solutions. The system needs to incorporate physical GPS devices together with predictive analytics and mobile application support for its development toward real-world implementation and system capacity development.

REFERENCES

- [1] J. Babu et al., "GPS Based Bus Tracking System," International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 13, 2021.

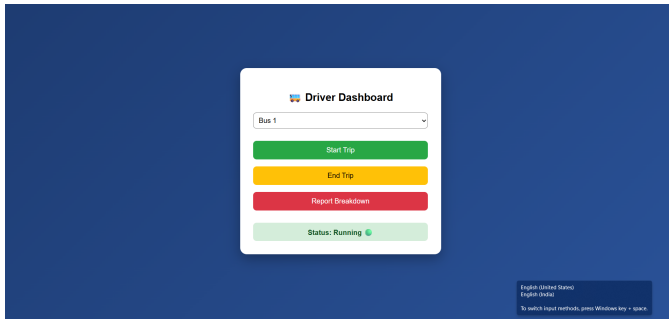


Fig. V.8. Driver dashboard during trip completion state.

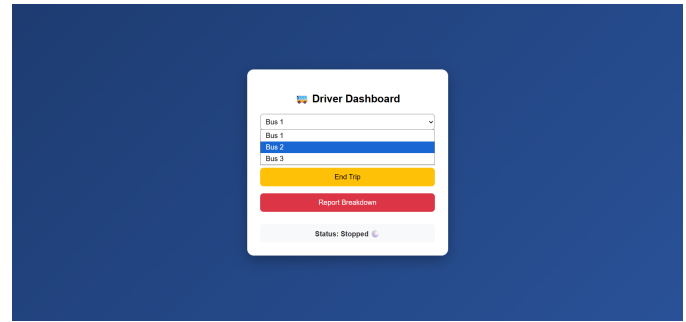


Fig. V.10. Initial driver control interface for trip management operations.

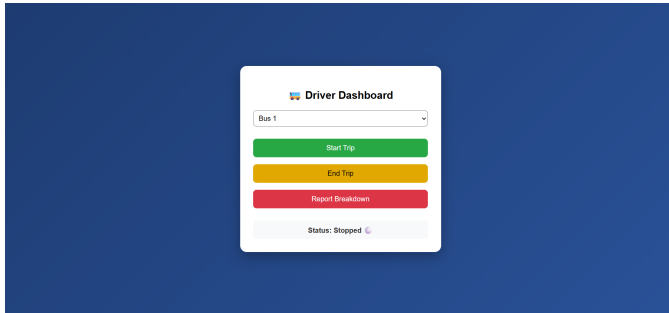


Fig. V.9. Driver dashboard during active transport operation.

- [20] G. Mintsis et al., "Applications of GPS Technology in Land Transportation," *European Journal of Operational Research*, vol. 152, pp. 399–409, 2004.

- [2] Y. H. A. Abdelhafed, "Improving Transportation Management by Real-Time Bus Tracking using GPS," *IJRASET*, 2025.
- [3] G. Gayathri et al., "Real Time Bus Tracking System," *International Journal of Advanced Research in Computer Science*, vol. 9, 2018.
- [4] P. S. Mane and V. D. Khairnar, "Analysis of Bus Tracking System Using GPS on Smartphones," *IOSR Journal of Computer Engineering*, vol. 16, no. 2, 2014.
- [5] R. Khopde et al., "Public Transport Tracking System," *International Education and Research Journal*, vol. 3, no. 1, 2017.
- [6] J. C. Ogbonna et al., "Real-Time Public Vehicle Mobile Tracking System Using GPS Technology," *International Journal of Computer Trends and Technology*, vol. 38, no. 1, pp. 71–80, 2016.
- [7] I. Madhur et al., "Smart Bus Tracking and Management System," *Iconic Research and Engineering Journals*, vol. 1, pp. 80–84, 2018.
- [8] S. Eken and A. Sayar, "A Smart Bus Tracking System Based on Location-Aware Services and QR Codes," in *Proc. IEEE INISTA*, 2014.
- [9] B. Thomas et al., "An Intelligent Location Tracker for Public Transport," *IJERT*, vol. 10, no. 4, 2021.
- [10] K. Gogri et al., "Real Time Bus Tracking System," *IJERT*, vol. 9, no. 6, 2020.
- [11] S. Eken and A. Sayar, "Smart Bus Tracking System Using QR Codes and GPS," in *IEEE INISTA Conference*, 2014.
- [12] H. Khaled et al., "Intelligent Transportation System Real-Time Tracking," *Qeios*, 2023.
- [13] A. Bessa et al., "RioBusData: Outlier Detection in Bus Routes," *arXiv preprint arXiv:1601.06128*, 2016.
- [14] K. Nguyen et al., "Los Angeles Metro Bus Data Analysis Using GPS Trajectory," *arXiv preprint arXiv:1909.00955*, 2019.
- [15] P. S. Mane and V. D. Khairnar, "Bus Tracking Using GPS Smartphones," *arXiv preprint arXiv:1407.0313*, 2014.
- [16] K. Ganesh et al., "Implementation of a Real Time Passenger Information System," *arXiv preprint arXiv:1206.0447*, 2012.
- [17] M. Z. Parvez et al., "A Theoretical Model of GSM-Based Vehicle Tracking System," in *Proc. ICECE*, 2010.
- [18] T. Le-Tien and V. P. The, "Routing and Tracking System for Mobile Vehicles," in *IEEE DELTA Conference*, 2010.
- [19] W. El-Medany et al., "A Cost Effective Real-Time Tracking System Using GPS/GPRS," in *IEEE Wireless and Mobile Communications*, 2010.